INTERNSHIP REPORT

TASK: 01

INTRODUCTION TO EMBEDDED SYSTEMS & IOT DEVICE DESIGN

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1. Introduction Section (Research)

O WHAT IS AN EMBEDDED SYSTEM?

An embedded system is a small computer that is built inside an electronic device to perform a specific task. It consists of both hardware and software that work together to control the operation of the device. Unlike a personal computer, which can perform many different tasks, an embedded system is designed for a particular function.

Embedded systems are widely used in our daily lives. Devices such as washing machines, microwave ovens, smart watches, traffic light controllers, automobiles, and smart home appliances all use embedded systems to perform their functions efficiently.

O REAL-WORLD APPLICATIONS

- Smart Homes – Smart lights, smart locks, security systems.
- Automobiles – Airbags, ABS, parking sensors, engine control.
- Healthcare – Digital thermometers, heart rate monitors, BP monitors.
- Consumer Electronics – Washing machines, microwave ovens, smart TVs.

O DIFFERENCE BETWEEN MICROCONTROLLER VS MICROPROCESSOR.

Parameter	Microprocessor	Microcontroller
Definition	Processing device computer systems.	Control device in an embedded system.
What is it?	Memory and I/O component are connected externally.	Memory and I/O output component ar present internally.
Circuit complexity	The circuit is complex due to external connection.	The circuit is less complex due to on chip resources
Efficiency	Microprocessors are not efficient.	Microcontrollers are efficient.
Number of registers	Microprocessors have less number of registers.	Microcontrollers have more number o registers.
Applications	Microprocessors are generally used in personal computers	Microcontrollers are generally used in washing machines, and air conditione

Conclusion

Both microcontrollers and microprocessors are important in electronic systems. A microcontroller is best suited for specific tasks in embedded systems, while a microprocessor is used for complex computing tasks in computers and laptops. Understanding their differences helps in selecting the right device for different applications.

2.COMPONENT STUDY

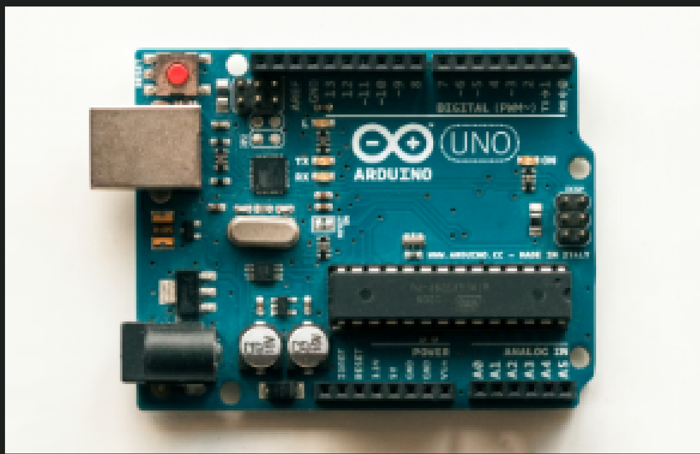
Research and explain the role of Microcontrollers (Arduino, ESP32, Raspberry Pi Pico).

1) Arduino UNO

Arduino UNO is a popular Microcontroller Board based on ATmega 328P microcontroller. It is widely used for learning electronics, Embedded Systems and IOT projects. The Aurdino UNO can read the data from the sensors and can control the devices like LED, Speaker, Display etc

Role of Arduino UNO

- Reads input from sensors.
- Processes the received data.
- Controls output devices.
- Used in automation and IoT projects.

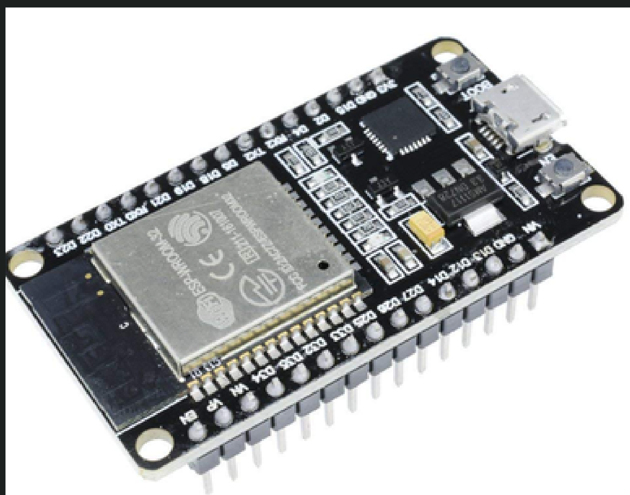


2) ESP 32

ESP32 is a Powerful microcontroller developed by Espressif Systems. It comes with built in Wi-Fi and bluetooth , making it ideal for IOT applications. It is commonly used in smart home systems, wireless monitoring and Automation projects.

Role of ESP 32

- Provides wireless communication
- Connects devices to the Internet
- Processes the sensor Data.
- Used in automation and IoT projects.

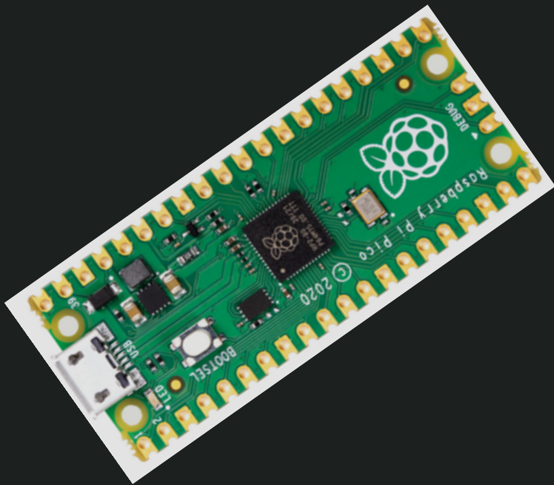


3) Raspberry Pi Pico

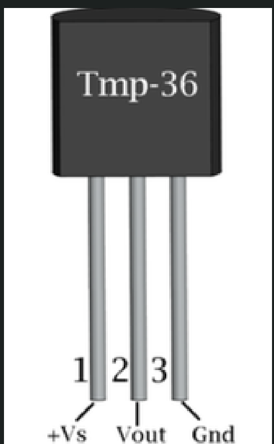
Raspberry Pi Pico is a compact and low-cost microcontroller board based on the RP2040 chip. It is suitable for embedded systems, Robotics applications

Role of Raspberry Pi Pico

- Controls electronic devices.
- Processes input from sensors.
- Supports automation projects.
- Used in robotics and embedded applications.



Basic sensors (Temperature sensor, Motion sensor, Light sensor).



Temperature sensor

PIR Motion sensor

LDR Light sensor

Temperature sensors, motion sensors, and light sensors help devices monitor conditions and respond automatically. These sensors improve the efficiency, accuracy, and intelligence of modern electronic systems used in homes, industries, healthcare, and IoT applications.

3. MINI PROJECT –
SMART SENSOR SYSTEM

1) Smart Temperature Monitor

Reads temperature values and shows alerts when crossing threshold.

Objective:

To monitor temperature and generate an alert when it exceeds a predefined threshold(35°C)

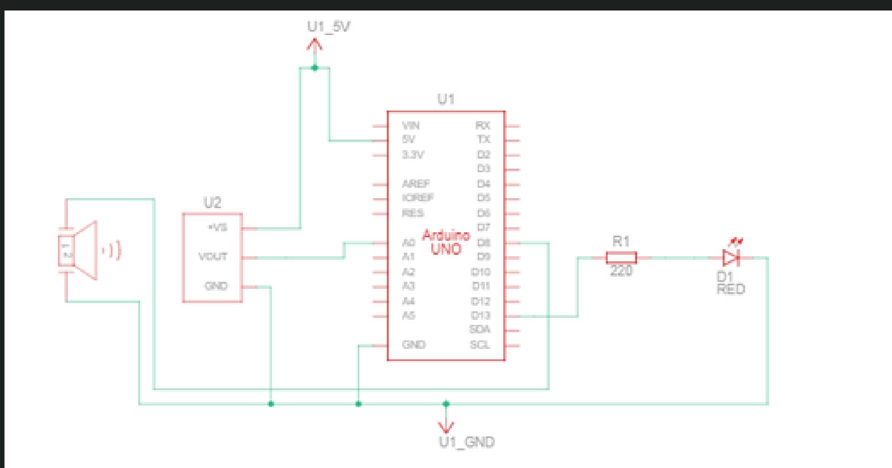
Components Used:

- Arduino UNO
- TMP36 Temperature Sensor
- LED
- Buzzer
- Resistor
- Jumper Wires

Software Used:

- TinkerCad

Circuit Diagram



Working Principle:

The TMP36 sensor measures temperature and sends the data to the Arduino. When the temperature exceeds 35°C, the Arduino turns ON the LED and buzzer to indicate a high-temperature condition. When the temperature drops below the threshold, the alert devices turn OFF.

Arduino Code:

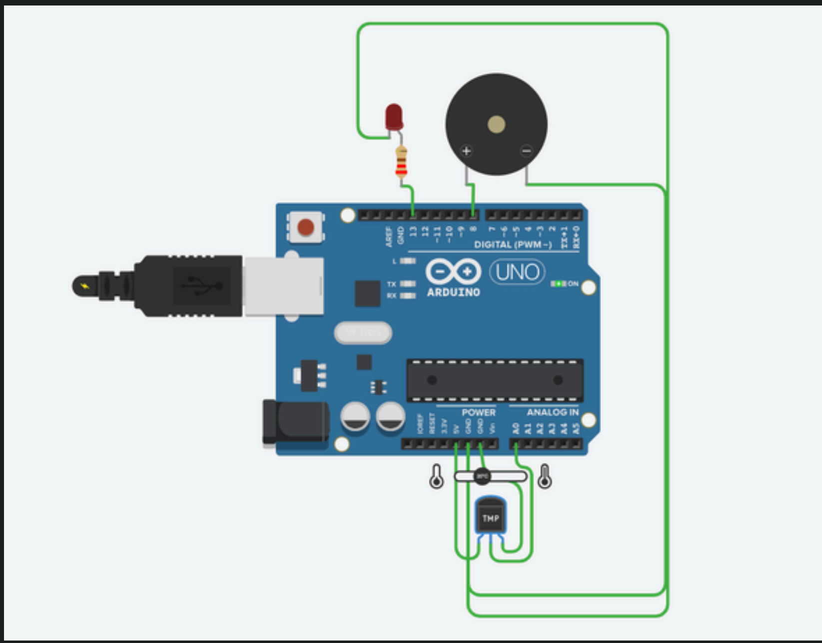
```
int tempPin = A0;      // Analog pin connected to TMP36 sensor
int ledPin = 13;       // Digital pin connected to LED
int buzzerPin = 8;     // Digital pin connected to Buzzer

void setup()
{
    pinMode(ledPin, OUTPUT);       // Set LED pin as output
    pinMode(buzzerPin, OUTPUT);    // Set Buzzer pin as output
    Serial.begin(9600);            // Start serial communication at 9600 baud
}

void loop()
{
    int sensorValue = analogRead(tempPin);         // Read analog value from TMP36
    float voltage = sensorValue * (5.0 / 1023.0); // Convert analog value to voltage
    float temperatureC = (voltage - 0.5) * 100;   // Convert voltage to temperature (°C)

    Serial.print("Temperature: ");     // Print label
    Serial.print(temperatureC);        // Print temperature value
    Serial.println(" °C");             // Print unit (Celsius)

    if (temperatureC > 35)
    {
        digitalWrite(ledPin, HIGH);     // If temp > 35°C → Turn ON LED
        digitalWrite(buzzerPin, HIGH); // If temp > 35°C → Turn ON Buzzer
    }
    else
    {
        digitalWrite(ledPin, LOW);      // If temp ≤ 35°C → Turn OFF LED
        digitalWrite(buzzerPin, LOW);  // If temp ≤ 35°C → Turn OFF Buzzer
    }

    delay(1000);          // Wait for 1 second before next reading
}
```

Output:

Case 1: When the Temperature is Below 35°C

Expected that - LED = OFF

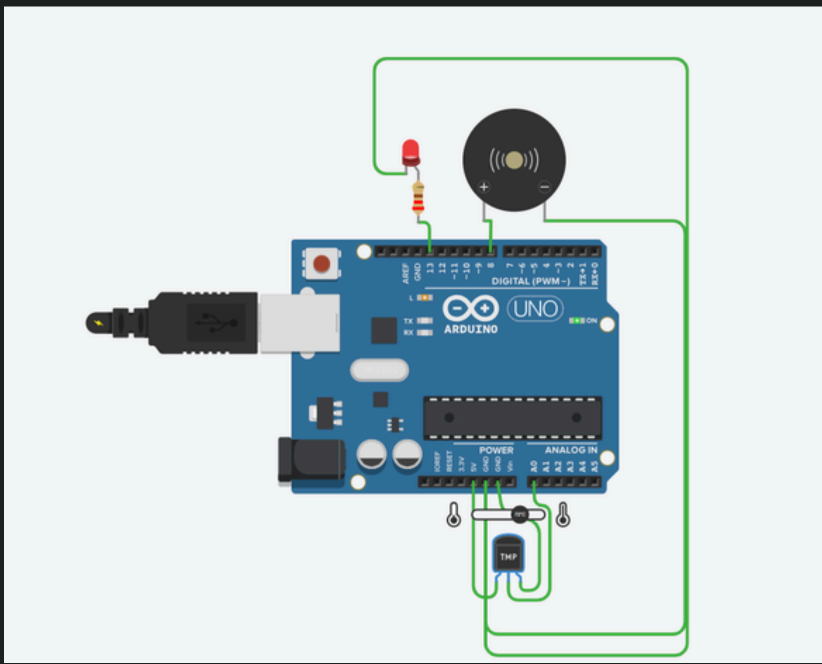
-Buzzer = OFF



Case 2: When the Temperature is Above 35°C

Expected that - LED = ON

-Buzzer = ON



Potential Real-World Applications

- Home Temperature Monitoring – Monitors room temperature and alerts users when it becomes too high.
- Server Rooms and Data Centers – Prevents overheating of electronic equipment by providing temperature alerts.
- Industrial Safety Systems – Monitors machine temperature and helps avoid damage due to excessive heat.
- Cold Storage Units – Ensures temperature remains within the required range for food and medicine storage.

Conclusion:

In this project, I built a Smart Temperature Monitor using Arduino and a temperature sensor. The LED and buzzer stayed OFF at normal temperature and turned ON when the temperature crossed 35°C. This helped me understand how sensors, Arduino, and simple alert systems work together in an embedded system.

THANK YOU !!!